



Apart from these challenges, more overlooked problems that lead to misjudgment and missed judgment are usually inevitable.

## State-of-the-art inspection machines

With all that said about AOIs, let us have a look at some of the latest AOI machines.

**Omron's S10 series.** It is a full colour system, offering a unique mix of general-purpose and application-specific solutions. S10 series enables flexible manufacturing, responds to shortages in manpower, and demands for higher quality production. Elimination of shadow and secondary reflection with 4-direction MPS projectors allows stable inspection of fine parts and enables visibility even at the connector solder joint. (<https://industrial.omron.eu/en/news-events/news/introducing-the-worlds-most-advanced-3d-aoi-system>)

**Mycronic's MYPro series 3D AOI system.** This newly designed user interface with intuitive programming guidance and self-optimising process control algorithms ensures easy implementation of high-performance 3D AOI for any type of production mix or class by operators of any experience level.

**Saki Corporation's 3Di-LS2-CASE.** It significantly reduces the number of false calls in the inspection of solder defects (non-wetting) that is difficult to achieve with conventional 3D-AOI technology, and dramatically improves the speed and accuracy of solder defect detection. (<https://metrology.news/next-generation-3d-aoi-system-with-artificial-intelligence-selected-by-denso>)

**CyberOptics's SQ3000 + multi-function system.** The SQ3000 + multi-function system for inspection and metrology not only conducts AOI and SPI but claims to deliver in-line, full coordinate measurement (CMM) data in seconds. The new SQ3000 + offers a combination of high accuracy and high speed with an even

| Defect type                       | AOI                     | AXI | ICT |
|-----------------------------------|-------------------------|-----|-----|
| <b>Soldering defects</b>          |                         |     |     |
| Open circuits                     | Yes                     | Yes | Yes |
| Solder bridges                    | Yes                     | Yes | Yes |
| Solder shorts                     | Yes                     | Yes | Yes |
| Insufficient solder               | Yes (not heel of joint) | Yes | No  |
| Solder void                       | No                      | Yes | No  |
| Excess solder                     | Yes                     | Yes | No  |
| Solder quality                    | No                      | Yes | No  |
| <b>Component defects</b>          |                         |     |     |
| Lifted lead                       | Yes                     | Yes | Yes |
| Missing component                 | Yes                     | Yes | Yes |
| Misaligned or misplaced component | Yes                     | Yes | Yes |
| Incorrect component value         | No                      | No  | Yes |
| Faulty component                  | No                      | No  | Yes |
| <b>BGA and CSP defects</b>        |                         |     |     |
| BGA shorts                        | No                      | Yes | Yes |
| BGA open circuit connections      | No                      | Yes | Yes |

(Source: Rayming PCB & Assembly)

higher resolution multi-reflection suppression sensor that inhibits reflection based distortions caused by shiny components and surfaces. (<https://www.businesswire.com/news/home/20211103005180/en/CyberOptics-Receives-Two-Awards-for-New-SQ3000%E2%84%A2-Multi-Function-and-WX3000%E2%84%A2-Systems-during-SMTAI>)

## Can AOI do it alone?

Automatic optical inspection works well in PCBs with clearly visible solder joints. However, many PCBs today are using other technologies, such as ball grid array (BGA) integrated circuits or chip scale packages (CSPs), where the solder joints are not visible. Such cases may need extended support with automated X-ray inspection (AXI) equipment that can check solder joints under components.

Manufacturers can consider ways to combine the inspection technologies rather than choosing between them, depending on the complexity of the PCB. Each of their advantages can be carefully balanced against cost to suit a particular production environment.

“AOI and AXI have been following the trend of most recent technologies to provide more performance for less cost. Another recent development is the portability of data between AXI and AOI systems from some suppliers. The ability to share libraries, inspection programs and statistical process control (SPC) data between AOI and AXI systems can greatly enhance machine usage and defect coverage. In addition, programming the systems of today is fast and intuitive. Operators are now able to quickly and easily create inspection programs and manage daily runtime operations with very little intervention. As a result, both high-mix and high-volume manufacturing lines can recognise immediate yield improvement and automated inspection results,” says Kevin Garcia, Applications Engineering Manager, Nordson YESTECH, an inspection equipment manufacturing company.

The advantages of current-day inspection systems over price and performance are becoming the driving force to handle dynamic manufacturing environments in giving successful yields. **EFY**